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JEE ADVANCED



Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

Sec: Sr. Elite(C-120, C-IPL, IPL-IC)

JEE ADV-2022_P2

Date: 08-10-23

Time: 02.30PM to 05.30PM

RPTA-08

Max. Marks:180

MATHEMATICS	:	Matrices & determinants
PHYSICS	:	Mechanical properties of matter Viscosity, surface tension Elasticity
CHEMISTRY	:	T-I: State of Matter T-II: Carboxylic acids & derivatives (acid halides, esters, amides, and hydrides, open chain & cyclic)

Name of the Student: _____

H.T. NO:

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JEE-ADVANCE-2022-P2-Model

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

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THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
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JEE MAIN 2023
SINGARAJU
VENKAT KOUNDINIA
IPL-12th Class
300
300
MARKS



RANK 1

JEE Advanced 2023
VAVILALA
CHIDULAS REDDY
IPL-12th Class
341
360
MARKS



RANK 1

NEET 2023
BORA VARUN
CHAKRAVARTHI
IPL-12th Class
720
720
MARKS



RANK 1

MATHEMATICS

Max Marks: 60

SECTION-I (INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct integer is entered;

Zero Marks: 0 If the question is unanswered;

Negative Marks: -1 In all other cases

01. Let A and B are two square matrices of order 3 such that $\det(A)=3$ and $\det(B)=2$, then the

value of $\det\left[\text{adj}\left(B^{-1}A^{-1}\right)^{-1}\right] = P$ then $\frac{P}{4} =$

02. Let three matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}; B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ then

$$T_r(A) + T_r\left(\frac{ABC}{2}\right) + T_r\left(\frac{A(BC)^2}{4}\right) + T_r\left(\frac{A(BC)^3}{8}\right) + \dots = \infty =$$

03. Let A,B,C be three square matrices of order '3' such that $A^2 + 2I = 0, \det(2C - A^2) = 32$ and satisfying $A^5 - 2A^3C + BA^2 - 2BC = 0$, then the absolute value of $[\det(B - A)]^2 =$

04. Let $A = [a_{ij}]$ be 3 x 3 invertible matrix. If determinant of A is 2, then the value of

$$\left| \left(\text{adj}(A^T) \right)^T \right| + \left| \left(\text{adj}A^{-1} \right)^{-1} \right| =$$

05. $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$, then f(x) is a polynomial of degree

06. If $l_1^2 + m_1^2 + n_1^2 = 1, l_2^2 + m_2^2 + n_2^2 = l_3^2 + m_3^2 + n_3^2 = 1$. and

$$l_1l_2 + m_1m_2 + n_1n_2 = l_2l_3 + m_2m_3 + n_2n_3 = l_1l_3 + m_1m_3 + n_1n_3 = 0, \text{ and } \Delta = \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}, \text{ then } |\Delta| =$$

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MAHARASHTRA



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07. If A, B, C are three square matrices of same order satisfying $A^2 = A^{-1}$ and let $B = A^{2^n}$ and $C = A^{2^{n-2}}$ ($n \in \mathbb{N}$) then $\det(B-C) =$ _____
08. If A is square matrix of order 3 such that $|A| = 2$, then $\left| (adj A^{-1})^{-1} \right|$ is _____

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

• This section contains SIX (06) questions.

• Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

• For each question, choose the option(s) corresponding to (all) the correct answer(s).

• Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks** +3 If all the four options are correct but ONLY three options are chosen;

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

09. Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ then correct statement is/are

A) $A^2 - 4A - 5I_3 = 0$

B) $A^{-1} = \frac{1}{5}(A - 4I_3)$

C) A^3 is not invertible

D) A^2 is invertible

10. If $A = \begin{bmatrix} 3 & 2 \\ 2 & \alpha \end{bmatrix}$; $B = \begin{bmatrix} 3 & 1 \\ 7 & 3 \end{bmatrix}$ and $\det(2A^9 B^{-1}) = -2$ then

A) $\alpha = 1$

B) $\alpha = -1$

C) $|A| = 1$

D) $|A| = -1$

11. For $\alpha, \beta, \gamma \in \mathbb{R}$, let $A = \begin{bmatrix} \alpha^2 & 6 & 8 \\ 3 & \beta^2 & 9 \\ 4 & 5 & \gamma^2 \end{bmatrix}$ and $B = \begin{bmatrix} 2\alpha & 3 & 5 \\ 2 & 2\beta & 6 \\ 1 & 4 & 2\gamma - 3 \end{bmatrix}$, if trace of A = Trace of B,

then

A) $\alpha^{-1} + \beta^{-1} + \gamma^{-1} = 3$

B) $\alpha = \beta = \gamma$

C) $2\alpha - 3\beta + \gamma = 0$

D) $\alpha + 2\beta + 3\gamma = 6$

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Sri Chaitanya
C-120 Class

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720**
MARKS



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12. If $\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$, then a factor of Δ is

A) $a + b + x$

B) $x^2 - (a - b)x + a^2 + b^2 + ab$

C) $x^2 + (a + b)x + a^2 + b^2 - ab$

D) $a + b - x$

13. If a, b, c are the roots of $x^3 + x^2 - 4 = 0$ and $A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ then

A) A is non singular

B) A is orthogonal

C) $AA^T = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

D) $A^{-1} = A^T$

14. Let $\alpha, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations $\alpha x + 2y = \lambda$, $3x - 2y = \mu$. Which of the following statement(s) is correct?

A) If $\alpha = -3$, then the system has infinitely many solutions for all values of λ and μ

B) If $\alpha \neq -3$, then the system has a unique solution for all values of λ and μ

C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $\alpha = -3$

D) If $\lambda = \mu \neq 0$, then the system has no solution for $\alpha = -3$

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SECTION – III (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

15. 'A' is an involuntary matrix given by $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ then the inverse of $\frac{A}{2}$ will be
- A) $2A$ B) $\frac{A^{-1}}{2}$ C) $\frac{A}{2}$ D) A^2
16. Let $A = [a_{ij}]$ be a 3×3 matrix, where $a_{ij} = \lim_{x \rightarrow 0} \frac{(1+ix)^{1/j} - 1}{x}$ for all $1 \leq i, j \leq 3$, then $A^2 =$ _____
- A) $4A$ B) $3A$ C) $2A$ D) A
17. If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & x(x-1)(x+1) \end{vmatrix}$ then $f(500)$ is equal to
- A) 0 B) 1 C) 500 D) -500
18. Let the homogeneous system of linear equations $px + y + z = 0, x + qy + z = 0$ and $x + y + rz = 0$, where $p, q, r \neq 1$, have a non-zero solution, then the value of $\frac{1}{1-p} + \frac{1}{1-q} + \frac{1}{1-r}$ is
- A) 0 B) -1 C) 1 D) 2

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PHYSICS

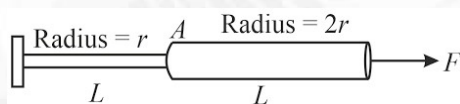
Max Marks: 60

SECTION-I (INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
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19. A metal cube of side 10 cm is subjected to a shearing stress of 10^6 N/m^2 . Calculate the modulus of rigidity (in $\times 10^8 \text{ Nm}^{-2}$) if the top of the cube is displaced by 0.05 cm with respect to its bottom.

20. Two steel rods of same length but radii r and $2r$ are connected together end to end and tied to a wall as shown.



The force stretches the combination by 10 mm. How far does the midpoint A move (in mm)?

21. A wire having a length L and cross-sectional area A is suspended at one of its ends from a ceiling. Density and Young's modulus of material of the wire are ρ and γ , respectively. Find its strain energy due to its own weight in μJ . (Given:

$$\frac{\rho^2 g^2 A L^3}{\gamma} = 12 \times 10^{-6} J).$$

22. Two soap bubbles in vacuum having radii 3 cm and 4cm respectively coalesce under isothermal conditions to form a single bubble. The radius of the new bubble is ____ cm.

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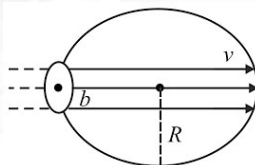


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23. A drop of liquid of radius $R = 10^{-2} m$ having surface tension $S = \frac{0.1}{4\pi} Nm^{-1}$ divides itself into K identical drops. In this process the total change in the surface energy $\Delta U = 10^{-3} J$. If $K = 10^\alpha$ then the value of α is _____.
24. n drops of water, each of radius $2mm$, fall through air at a terminal velocity of $8cms^{-1}$. If they coalesce to form a single drop, then the terminal velocity of the combined drop is $32 cms^{-1}$. Find n .
25. A soap film is made by dropping a circular frame of radius b in soap solution. A bubble is formed by blowing air with speed v in the form of a cylinder. Air stops after striking surface of soap bubble. Density of air is ρ . The radius of the bubble is $R > b$. The radius R of the bubble when the soap bubble separates from the ring is $\frac{\beta T}{\rho v^2}$. Find the value of β . (where, T is the surface tension of the liquid)
- 
26. The capillaries of radius r and $2r$ and lengths l and $2l$ respectively are joined in series. The streamline flow is maintained in capillaries. The pressure difference across first and second capillary is P_1 and P_2 respectively. Find P_1 / P_2 .

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions.
 - Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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- $$v = v_0 \left(1 - \frac{r^2}{R^2} \right), \text{ where } r \text{ is the radial distance of that location from the axis of the pipe and}$$

$\pi/2 \times 10^{-2} \text{ m}^3 \text{ s}^{-1}$ and the coefficient of viscosity of the liquid is 0.75 N s m^{-2}

- A) Magnitude of viscous force per unit area at $r=4$ cm is 6 N/m^2
B) Maximum velocity of flow of liquid is 1 m/s
C) Viscous force per unit area increases with radial distance
D) Viscous force per unit area decreases with radial distance

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2023**

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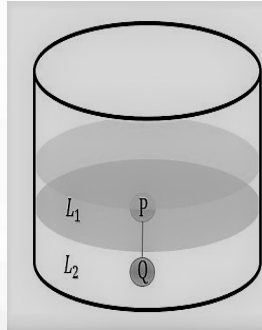
**NEET
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720
720



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31. Two spheres P and Q of equal radii have densities ρ_1 and ρ_2 , respectively. The spheres are connected by a massless string and placed in liquids L_1 and L_2 of densities σ_1 and σ_2 and viscosities η_1 and η_2 , respectively. They float in equilibrium with the sphere P in L_1 and sphere Q in L_2 and the string being taut (see figure). If sphere P alone in L_2 has terminal velocity $\vec{V}_P$ and Q alone in L_1 has terminal velocity $\vec{V}_Q$, then



- A) $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_1}{\eta_2}$ B) $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_2}{\eta_1}$ C) $\vec{V}_P \cdot \vec{V}_Q > 0$ D) $\vec{V}_P \cdot \vec{V}_Q < 0$
32. Eight identical droplets, each falling under gravity in the earth's atmosphere with terminal velocity, v , combine to form a single drop. The terminal velocity of single drop and drag force on it are calculated. Find the correct statement from the following.
- A) Final terminal velocity is $4v$ B) Final terminal velocity is $8v$
 C) Drag force is increased to four times D) Drag force is increased to eight times

SECTION – III (SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

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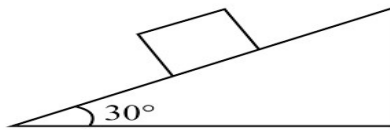
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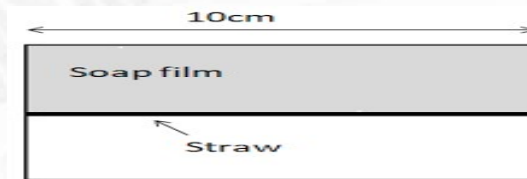


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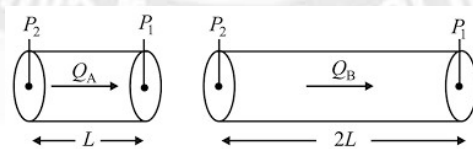
33. A rain drop starts falling from a height of 2 km. It falls with a continuously decreasing acceleration and attains its terminal velocity at a height of 1 km. The ratio of the work done by the gravitational force in the first half and the second half of the drops journey is
 A) 1:1 and the times of fall of the drop in the two halves is $a:1$ (where $a>1$)
 B) 1:1 and the times of fall of the drop in the two halves is $a:1$ (where $a<1$)
 C) $a:1$ (where $a>1$) and the times of fall of the drop in the two halves is 1:1.
 D) $a:1$ (where $a<1$) and the times of fall of the drop in the two halves is 1:1.
34. A cubical block (of side 2 m) of mass 20kg slides on inclined plane lubricated with the oil of viscosity $\eta = 10^{-1}$ poise with constant velocity of 10 m/s ($g = 10 \text{ m/s}^2$). The thickness of layer of liquid is (10^{-1} poise $= 10^{-2} \text{ N s m}^{-2}$)



- A) 2.5 mm B) 4 mm C) 6 mm D) 5.5 mm
35. A soap film of surface tension $3 \times 10^{-2} \text{ N m}^{-1}$ formed in a rectangular frame can support a straw as shown in figure. If $g = 10 \text{ m/s}^2$, the mass of the straw is



- A) 0.006 g B) 0.06g C) 0.6g D) 6 g
36. A viscous fluid is flowing through two horizontal pipes. The pressure difference $P_2 - P_1$ between the ends of each pipe is the same. The pipes have the same radius, although one is twice as long as the other. How does the volume flow rate Q_B in the longer pipe compare with the rate Q_A in the shorter pipe?



- A) Q_B is the same as Q_A B) Q_B is twice as large as Q_A
 C) Q_B is four times as large as Q_A D) Q_B is one-half as large as Q_A

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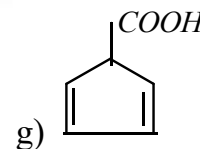
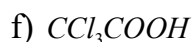
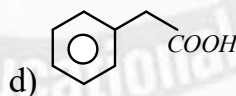
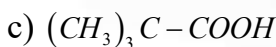
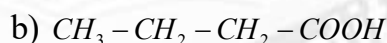
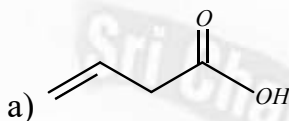
CHEMISTRY

Max Marks: 60

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

37. Two moles of Ammonia gas are confined to a five litre flask at 27°C . The pressure of, the gas is $(X+1)$ atmospheres according to the Vander Waal's equation. (For NH_3 $a = 6.25 \text{ atm lit}^2 \text{ mol}^{-2}$ and $b = 0.037 \text{ lit mol}^{-1}$). Find the value of 'X'
38. A 10L flask contains 41.4g of a mixture of gases C_xH_8 and C_xH_{12} . The total pressure of 44°C in the flask is 1.56 atm. Analysis revealed that the gas mixture has 87% total carbon and 13% total H. The value of 'X'.
39. A flask has 10 gas particles, out of which 4 particles are moving of 7 ms^{-1} and the remaining are moving at the same speed of $X \text{ ms}^{-1}$. If the r.m.s speed of the gas is 5 ms^{-1} . What is the value of 'X'?
40. If an ideal gas of 100K is heated to 109K, in a rigid container, the pressure increases by X%. The value of 'X'.
41. An organic compound. A (mol.wt=180) is acylated with CH_3COCl to get acylated compound of mol wt. 390. What is the number of amino groups present per molecule of the compound A.
42. The number of compounds among the following which are more reactive than acetic acid towards decarboxylation by Soda-lime is/are



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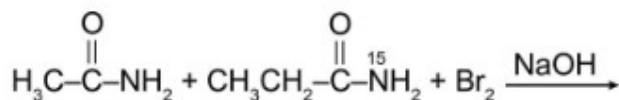
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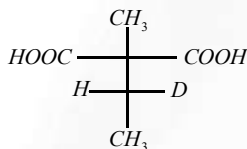

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43. In the following Hofmann's bromamide reaction, how many different amines would be formed?



44. find the total number of different organic and inorganic products obtained when

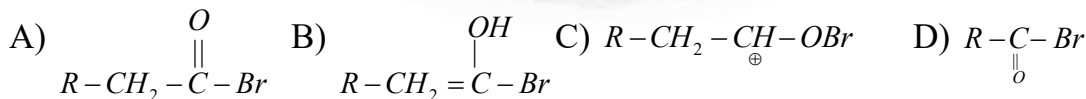


is heated strongly.

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

- This section contains SIX (06) questions.
- Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

45. An organic compound $\text{A}(\text{C}_5\text{H}_{10}\text{O}_2)$ reacts with Br_2 in presence of phosphorous to produce B which can be resolved into enantiomers. B on dehydrobromination yields C ($\text{C}_5\text{H}_8\text{O}_2$) which does not show stereoisomerism. C on decarboxylation gives an alkene which on further ozonolysis gives D and E. D gives Fehling's test but E does not. The correct identity of these compounds are
- A) A is $(\text{CH}_3)_2\text{CHCH}_2\text{COOH}$ B) B is $(\text{CH}_3)_2\text{CHCHBrCOOH}$
 C) C is $(\text{CH}_3)_2\text{C}=\text{CHCOOH}$ D) E is acetone
46. The structure of the intermediate formed in the HVZ reaction of Bromine with RCH_2COOH .



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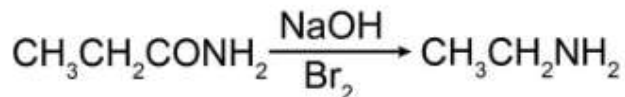
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47. Consider the following reaction



What is/are true regarding the above reaction?

- A) The reaction proceeds through a free radical intermediate.
 B) The reaction involves $\text{CH}_3\text{CH}_2\text{NCO}$ intermediate.
 C) The reaction involves an intramolecular alkyl shift.
 D) If a (2°) amide is taken, N-alkyl amine is formed.
48. Indicate the correct statement for equal volumes of $\text{N}_{2(g)}$ and $\text{CO}_{2(g)}$ of 25°C and 1 atm.
- A) The average translational KE per molecule is the same for N_2 and CO_2 .
 B) The rms speed is same for both N_2 and CO_2
 C) The density of N_2 is less than that of CO_2
 D) The total translational KE of both N_2 and CO_2 is the same
49. U_{av} , U_{rms} and U_{mp} are average, root mean square, and most probable speeds of the gas respectively at constant temperature T. The molar mass of the gas is M. Then
- A) no molecules can have a speed greater than $\sqrt{2} U_{rms}$
 B) no molecule can have a speed less than $U_{mp} / \sqrt{2}$
 C) $U_{mp} < U_{av} < U_{rms}$
 D) The Kinetic energy of a mole of gas is $\frac{3}{4} M U_{mp}^2$
50. the time taken for effusion of 32ml of O_2 will be nearly the same as the time taken for effusion under identical conditions of
- A) 64 ml of H_2 B) 50 ml of N_2 C) 27.3 ml of CO_2 D) 22.62 ml of SO_2

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SECTION – III

(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

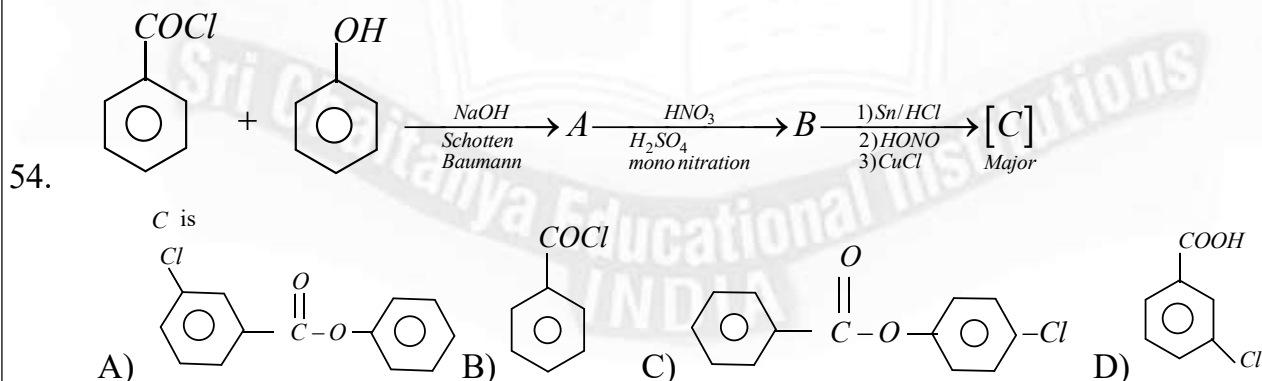
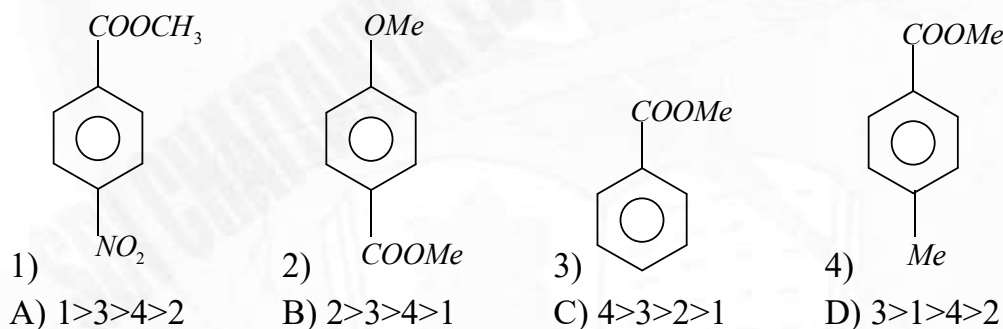
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

51. Identify the incorrect statement in the following?
- A) Greater the viscosity more slowly the liquid flows..
 B) Surface tension of Ethyl alcohol at 500 K is less than that at 500°C
 C) Lowest energy state of liquid drop is spherical shape.
 D) Standard boiling point of water is 99°C and its normal boiling point is 100°C.
52. Oxygen diffuses from two different vessels A and B under identical conditions but A has circular orifice while B has square orifice of length equal to the radius of the orifice of vessel A. The ratio of rate of diffusion of the gas from vessel A to that from vessel B is
- A) $2 : \pi$ B) $4 : \pi$ C) $\pi : 4$ D) π
53. The decreasing order of ease of hydrolysis of the following ester is



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BELOW 100
All India Open
Category Ranks

32

BELOW 1000
All India Open
Category Ranks

181

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Count

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BELOW 1000
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